

NCIE-014 Skeletal v0.1 → Full v1.0

Expansion Verification Report

Database & Storage Engineering Specification — IN DEVELOPMENT / FOR HUMAN REVIEW OF FULL EXPANSION

1. Title Verification (Source-Confirmed at Skeletal Stage)

The approved NCIE-014 Skeletal v0.1 Document Governance block already records a source-confirmed title ('NCIE-014 — Database & Storage Engineering Specification') and source-confirmed suite purpose ('Gives Codex implementation-level persistence design'). This was checked against the NCIE Documentation Suite Master Controlled Register v2.0, Section 3, NCIE-014 entry, and found to match exactly in both title wording and Primary Purpose framing. Unlike the NCIE-013 v1.0 edition, which required a pre-approval surgical correction after its original build incorrectly stated no register was supplied, no title-reconciliation discrepancy exists for NCIE-014 and no correction was necessary. This finding is SOURCE-VERIFIED against the register text itself, not inferred from the document's naming convention or conversational context.

2. Scope Verification

NCIE-014 v1.0 is a controlled full expansion of the approved NCIE-014 Skeletal v0.1: every approved chapter purpose, Salient Things to Note, principal component, upstream dependency/authority and Human Review focus is elaborated into canonical-to-physical mapping governance, relational/analytical/object/search/vector/graph/time-series persistence design, registry/memory/Evidence persistence, physical keys/indexing/partitioning/replication, caching, classification/residency/encryption enforcement, retention/backup/DR/migration/reconciliation engineering, database security, storage observability/performance/cost and persistence testing, without inventing database/search/vector/graph/cache/backup technology selections, RPO/RTO, replica/partition counts, cache TTLs, retention periods, backup frequency, lag thresholds, storage SLOs or capacity thresholds absent source authority.

3. 34-Chapter Integrity Verification

A programmatic scan of the final document confirms exactly 34 sequential 'NCIE-014 — Chapter N Expansion' labels, N = 1..34, each present exactly once, in the approved order, with no chapter added, removed, merged, split or renamed relative to the approved Skeletal v0.1 structure.

4. Salient Things to Note Hard-Gate Verification

Per directive item 75 — continuing the discipline established after the NCIE-009 Chapter-27 omission — every chapter must retain an explicit '<N>.0 Salient Things to Note' subsection. This build enforced the requirement structurally: the chapter-rendering helper (lib.py's add_chapter function) raises a build-time assertion error if a chapter's salient content is omitted, so the document cannot be assembled with a missing subsection. A programmatic scan of the final assembled document, rerun independently rather than inferred from the build script succeeding, confirms an explicit Salient Things to Note heading present for all 34 chapters (N = 1 through 34), with zero missing.

5. Skeleton Fidelity Verification

Each of the 34 chapters was authored directly from its approved skeletal Chapter Purpose, Salient Things to Note, Principal Components, Upstream Dependencies/Authority and Human Review Focus (see the accompanying Change Register for the per-chapter mapping). No approved chapter concept was omitted, and none was conceptually inverted.

6. No-Technology-Invention Audit (Directive Item 8)

A full-text scan for unsourced relational, analytical, object, search, vector, graph, time-series, cache, backup or archival product selections found none — every store-technology decision (Chapters 4-11, 17, 23) is stated as a Human Review item (e.g. HR14-4-1, HR14-7-1, HR14-8-1, HR14-9-1, HR14-10-1) rather than a mandated product, consistent with directive item 8.

7. No-Numeric-Invention Audit (Directive Item 9)

A full-text scan for invented RPO/RTO, replica counts, partition counts, cache TTLs, retention periods, backup frequency, lag thresholds, storage SLOs, performance targets or capacity thresholds found none — Chapter 17 defers partition/shard counts (HR14-17-1), Chapter 19 defers cache TTL/invalidation specifics (HR14-19-1), Chapter 22 defers retention periods (HR14-22-1), Chapter 23 defers RPO/RTO (HR14-23-1) and Chapter 29 defers SLO/capacity thresholds (HR14-29-1) — every concrete number is a named Human Review item, never an invented default.

8. Canonical-Drift Audit (Directive Item 64)

A full-text scan for physical schema/index/cache/projection wording that changes NCIE-003 semantics found none — Chapter 3's mapping-register discipline (Table 5) and the repeated PHYSICAL SCHEMA ≠ CANONICAL SEMANTIC MODEL invariant (§0.3) consistently frame every physical structure as a realization of, never a redefinition of, its NCIE-003 canonical source.

9. Current-State Audit (Directive Item 65)

A full-text scan for replicas/caches/backups/restored copies being treated as authoritative current authorization/eligibility/state without reconciliation found none — Chapter 18's Authoritative-Read Matrix (Table 21), Chapter 19's cache-boundary discipline and Chapter 23's restore-reconciliation requirement (§23.2, Table 29) consistently require reconciliation before any non-primary copy is treated as authoritative.

10. Historical-Integrity Audit (Directive Item 66)

Deletion (Chapter 22), retention/archival (Chapter 22), migration (Chapters 25-26) and repair/restore (Chapters 23, 27) were each verified to preserve required Evidence/audit/provenance/historical state — tombstones are used where governed history must remain (§22.2), and repair/restore both require post-action verification (§23.2, §27.2) rather than silent overwrite.

11. Security/DBA Audit (Directive Item 67)

A full-text scan confirms DB administration (Chapter 28), encryption (Chapter 21), locality (Chapter 20) and partitioning (Chapter 17) are consistently represented as implementation mechanisms, never as content authority, authorization, residency compliance or a security boundary by themselves — each chapter restates its own invariant (DATABASE ADMINISTRATION ≠ CONTENT AUTHORITY; ENCRYPTION ≠ AUTHORIZATION; DATA LOCALITY ≠ RESIDENCY COMPLIANCE BY ITSELF; PARTITIONING ≠ SECURITY/TENANT BOUNDARY BY ITSELF).

12. Migration Audit (Directive Item 68)

Chapters 25 and 26 were verified to distinguish technical migration success from semantic reconciliation/acceptance — §25.1 requires upstream canonical governance approval for breaking semantic change, and §26.1-26.2 state explicitly that MIGRATION SUCCESS ≠ RECONCILIATION COMPLETE, with exception handling and provenance preserved through cutover.

13. Implementation-Claims Audit (Directive Item 71)

A full-text scan for language claiming stores, migrations, backups, DR, encryption, indexes or recovery controls are implemented, validated or production-ready found none — every chapter's status line reads 'IN DEVELOPMENT — FOR HUMAN REVIEW' and Chapter 32's test-status framework (§32.2) marks every test class SPECIFIED only, per directive item 63's six-state distinction (TEST SPECIFIED, IMPLEMENTATION EVIDENCE AVAILABLE, TEST EXECUTED, TEST PASSED, RECOVERY/MIGRATION VALIDATED, HUMAN/PRODUCTION ACCEPTANCE).

14. Adversarial Review (Directive Item 72)

The document was reviewed from data-architecture, DBA, distributed-systems, search/vector/graph, security/privacy, Evidence/provenance, migration/recovery and operations perspectives. Corrections applied during review: ensured every store-allocation and technology-selection statement routes to a Human Review item rather than an implied default; ensured the Recovery/Reconciliation Matrix (Table 29) explicitly covers migration and repair scenarios, not only backup/DR, closing a gap an operations-only reading would have missed. No defect exposing private chain-of-thought was found or is reported.

15. Skeleton Fidelity Audit

All 34 approved Chapter Purposes, Salient Things to Note, Principal Components, dependencies and Human Review foci were verified represented and not inverted, per the mapping in the accompanying Change Register.

16. 34-Chapter Integrity Audit (Directive Item 74)

Programmatically re-verified: exactly 34 sequential chapters, no missing/duplicate/added chapter and no unintended renaming.

17. Salient Things to Note Audit (Directive Item 75)

Programmatically re-verified: an explicit Salient Things to Note subsection is present in all 34 chapters.

18. Human Review Register Mechanical Reconciliation

Chapter 33's register was recomputed programmatically by scanning the Human Review Focus bullet text already rendered in Chapters 1-32, plus HR14-33-1 (Chapter 33's own item) and HR14-34-1 (Chapter 34's own item) — both defined once and included in the register-construction list from the start of the build, a design choice made specifically to avoid repeating the register-reconciliation defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Result: 34 total items, 29 Blocking / 5 Non-Blocking, 0 duplicate Review IDs, 0 orphan Human Review items. Classification breakdown: 12 Institutional Confirmation Required, 10 Security/Sovereignty Confirmation Required, 9 Engineering/Infrastructure Discovery Required, 2 Legal/Policy Confirmation Required, 1 Resolved by Upstream Baseline.

19. Upstream Human Review Deduplication (Directive Item 62)

Every NCIE-014 Human Review item was individually checked against the existing NCIE-008–013 Human Review registers. None was found to restate an identical institutional, security, Evidence, integration, UX or domain-business question — each concerns a distinct persistence-engineering decision those upstream documents raise the need for but do not themselves resolve. No cross-reference deduplication was required for this edition.

20. Blocking Scope Discipline

Each Blocking item in the Chapter 33 register states precisely which technology selection, numeric threshold, retention/residency rule or ownership assignment it blocks, rather than implying the whole document or the whole approved upstream chain is blocked. Non-Blocking items (Chapters 9, 15, 16, 30, and the resolved Chapter 34 item) are explicitly marked as not gating the corresponding chapter's persistence operation in the interim.

21. Canonical-to-Physical Mapping Matrix Verification (Directive Item 56)

Table 5 (Chapter 3) was verified to map every canonical element class (entity, field, identifier, relationship, type/constraint, derived field) to a physical structure class, mapping type and version-linkage mechanism, with semantic authority explicitly retained by NCIE-003 in every row.

22. Physical-Store Allocation Matrix Verification (Directive Item 55)

Table 2 (Chapter 1) was verified to map every canonical/logical object class introduced later in the document (transactional entities, analytical aggregates, files/objects, search-indexed content, embeddings, graph projections, time-series, registry/Memory state, Evidence/audit records) to a physical role, authority source, security class and history requirement, without creating semantic ownership at the physical layer.

23. Authoritative-Read, Residency-Copy, Retention/Disposition and Recovery/Reconciliation Matrix Verification (Directive Items 57-60)

Table 21 (Chapter 18) was verified to state an explicit read requirement, staleness indicator and reconciliation path for six workload classes. Table 24 (Chapter 20) was verified to cover all seven copy types (primary, replica, index, cache, backup, DR, archive). Table 27 (Chapter 22) was verified to map five logical retention/disposition states to physical obligations without inventing a period. Table 29 (Chapter 23) was verified to cover five recovery/migration/repair scenarios with restore source, reconciliation, authorization refresh, integrity check and acceptance evidence stated for each.

24. Character-Encoding Integrity Check

Following the same build pipeline used for the whole NCIE series (UTF-8 source fragments executed via a Python `exec()` call with an explicit `encoding='utf-8'` argument, a fix first required during the NCIE-008 build), a full-document scan of the final NCIE-014 document confirmed zero mojibake occurrences across all paragraphs and tables.

25. Visual and Structural Verification

The complete document was opened in Microsoft Word, its Table of Contents field (a live Word field, not static text) was updated and confirmed to populate with correct chapter/section titles and real page numbers, repaginated to 60 pages, saved, and rendered to PDF. Independent text extraction from the rendered PDF confirms the cover occupies page 1 alone and Document Governance begins on page 2 (the manual page break), with the running header and footer correctly reading NCIE-014 branding. No table clipping, overlapping text, or broken table was observed across the six consolidated matrices, the Chapter 33 Human Review Register tables, or the Chapter 34 traceability/handover tables.

26. Final Human Review Extraction (Authoritative)

Performed directly against the final specification file by independently scanning every paragraph and table cell for the pattern `HR14-<chapter>-<n>`: TOTAL IDS FOUND = 34. UNIQUE IDS = 34. DUPLICATES = 0. OPEN = 33. RESOLVED = 1 (HR14-34-1, Resolved by Upstream Baseline). BLOCKING = 29. NON-BLOCKING = 5. CLASSIFICATION TOTALS: Institutional Confirmation Required = 12; Security/Sovereignty Confirmation Required = 10; Engineering/Infrastructure Discovery Required = 9; Legal/Policy Confirmation Required = 2; Resolved by Upstream Baseline = 1. ORPHAN IDS = 0 — every one of the 34 unique IDs found anywhere in the document also appears as a row in the Chapter 33 Complete Register table, which itself contains exactly 34 rows with no duplicate Review ID.

27. Three-Artifact Consistency Check

The NCIE-014 v1.0 main document, this Verification Report and the accompanying Full-Expansion Change Register agree exactly on: Formal title 'NCIE-014 — Database & Storage Engineering Specification' (source-confirmed at skeletal stage, verified against Master Controlled Register v2.0, no correction required); Source Version v0.1 (Skeletal) → Target Version v1.0 (Full); Authority chain approved NCIE-001 v1.3 through approved NCIE-013 v1.0, plus the approved NCIE-014 Skeletal v0.1; Status IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION (not Approved); 34 chapters preserved in order (0 missing, 0 duplicate, 0 added, 0 unauthorized renaming); Salient Things to Note hard gate = 34/34 explicit subsections present; Human Review Register TOTAL = 34, BLOCKING = 29, NON-BLOCKING = 5, classification totals as in §26 above; orphan Human Review IDs = 0; duplicate Human Review IDs = 0; persistence test-

class status = 9 tests specified, none executed, passed, or recovery/migration-validated; database/storage invariants = 21 NCIE-014-specific plus cross-referenced upstream invariants; downstream handover = NCIE-015/016/017 each receive a defined interface without technical preemption. No mismatch was found between the three artifacts.

28. Final Readiness Disposition

NCIE-014 — Database & Storage Engineering Specification v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed technology selection, numeric threshold, retention/residency rule and ownership assignment deferred in Chapters 1-32 remains a proposal pending resolution of the Chapter 33 Human Review Register; every §32.1 test remains SPECIFIED, not executed, passed, or recovery/migration-validated, pending NCIE-016. This document is NOT marked Approved.